PROJECT PROPOSAL · SOFTWARE ENGINEERING LAB - United International University

Smart Queue Management System

Digital tokens, live queue tracking and smart alerts, so nobody waits in line.

Name: Mohammad Saiful Islam Fahim· Faculty: Faysal Ahmed · Tools: Git, Jira, Node.js, MySQL

Problem statement

Hospital, banks and university offices still rely on physical lines and paper tokens.

No visibility

People cannot see their position or how long the wait will be.

Time wasted

Visitors must stay on site, even when their turn is far away.

Poor management

Staff have no data on peak hours, no-shows or service times.

Proposed idea and objectives

A web app where visitors take a digital token, watch their position update live, and get notified before their turn.

- Replace physical lines with real-time digital queues
- Estimate wait time from past service data
- Alert users by email when their turn is near
- Give staff and admins tools and analytics

Key features and APIs

API	Used for
Google OAuth	One-click login and visit history
Maps API	Branch map and a "Leave now" travel-time alert
Email API	Token confirmation and turn-near alerts
QR Code API	Token QR for staff check-in and no-show control

Also: staff panel, public display board, priority tokens, appointments, analytics, English and Bangla.

Method and architecture

Frontend

React web app (PWA) for customers, staff and admin.

Backend

Node.js and Express REST API, with Socket.IO for live updates.

Database

MySQL with transactions and row locking to stop double-calling a token.

Process: Agile Scrum in five sprints. Jira for epics, stories and bugs; Git feature branches, pull requests and tagged releases; GitHub Actions for tests and deployment.

Plan and timeline

Sprint 1

Setup, auth, schema, token API

Sprint 2

Staff panel, call-next logic

Sprint 3

WebSockets, display board, email

Sprint 4

Maps, QR check-in, appointments

Sprint 5

Analytics, testing, CI/CD, deploy

Deliverables: SRS, ER and use case diagrams, Jira board, working app, test report.

Expected outcome

- Shorter, more predictable waiting for visitors
- Better workload control and data for staff
- A deployed, tested system that shows real Git and Jira practice

Questions?

Thank you

